A062 — SST Microgravity Force-Gradient Falsifier

v0.1.0 Scientific Protocol and Standalone Numerical Report

SST-Workbench — Candidate Model Analysis

12 October 2026

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	88568689617728e36a2d7a9e9ca78a6257becbb2f10d189899e50fc3ca4e5e5f
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	f4faef6264e7a9486d25cf629846c9cf7207de5415519b77a1d0e4eb93403801

Gate ledger

Gate	Status	Question
G0	PASS	Frozen protocol, local provenance and runtime integrity verified?
G1	PASS	Are the published experimental benchmark and source files admitted?
G2	PASS	Are Python FP64 numerical quadrature and dimensional tests qualified?
G3	FAIL	Does the registered candidate satisfy mass and far-field closure with fixed paran
G4	UNRESOLVED	Are withheld independent experimental confirmation data available and qualifie
G5	PASS	Did actual C++ FP64 source quadrature and Python parity execute?
G6	NOT_APPLICABLE	Did an actual supported GPU screening kernel run with proven source and pre
G7	NOT_APPLICABLE	Was independent multi-source cross-replication achieved?
G8	UNRESOLVED	Was a genuine PKLSA vortex-knot source used with a physically derived force
G9	DEFERRED	Was reveal integrity executed and authorized?

Backend manifest


```
{
  "cpp_fp64": {
    "authority": "C++ FP64 solver-specific parity only; not v1.0.7 backend certification",
    "compiler": "g++",
    "max_relative_error": 6.7091685042337876e-12,
    "ran": true,
    "requested": true
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
```

```
  "scientific_claim": "restricted Gaussian Euler pressure source only; not full SST",
  "sycl_gpu": {
    "authority": "SCREENING_ONLY",
    "ran": false,
    "requested": false
  }
}
```


Source manifest


```
{
  "manifest_sha256": "a3e27cd441b306914a12aa4231375487eb9619332cb29657619fc0e9957d37e4",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "Nature Nanotechnology doi:10.1038/s41565-026-02298-8",
      "evidence_class": "experimental",
      "exists": true,
      "independence_family": "Zhu2026 single experiment",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_sst_mic
      "provenance_family": "Zhu2026",
      "registry_hint": {
        "catalog_id": "Nature Nanotechnology doi:10.1038/s41565-026-02298-8",
        "hits": []
      },
      "role": "public_benchmark_only",
      "sha256": "47f0e0abcae5f06e1087faaf1d8b4b9f4b7771638164fc2e529d61234494320c",
      "source_id": "S0",
      "version": "2026 article PDF user supplied"
    },
    {
      "catalog_id": "SST candidate analytic Gaussian Euler-Poisson",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "analytic_theory",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_sst_mic
      "provenance_family": "local_generated",
      "registry_hint": {
        "catalog_id": "SST candidate analytic Gaussian Euler-Poisson",
        "hits": []
      },
      "role": "model",
      "sha256": "f005abc6d5484b8d519423e2837e3a1ebf7039c8388b66d7faccce5d9e9f6748",
      "source_id": "S1",
      "version": "A062 v0.1.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


Status: restricted model diagnostics, not a test of SST canon as a whole. The target-bearing data of the published article are already public and are therefore *not* scientifically blind. The science-only run is separate from the canonical frozen v1.0.7 pipeline; no framework certification or real reveal is claimed. Protocol SHA-256 (prefix): `5dd2f0d5ce2df29b`.

1 Question, provenance and expected falsifiers

Can an explicit incompressible, inviscid, localized azimuthal Gaussian vortex yield both a mass-equivalent source and a gravitational far-field force gradient with $F \propto r^{-2}$, without injecting G or fitting its amplitude? The mechanical instrument uses a 100-g test mass and a movable 1.2-g source at 4.2 cm separation, with 1.4 μm modulation. These are public rounded values reported by Zhu et al. [1]. The tested candidate is *not* a PKLSA knot and the surrogate source width is a chosen coarse-graining scale $a = 3\,\mathrm{mm}$, not the canonical r_c.

Null and alternative. Under H0 the restricted Gaussian mass/pressure closure fails at least one pre-registered scaling/mass test. Under H1 this closure passes both the mass ratio (within 10%) and the inverse-square slope (2 ± 0.2), on both source orientations, with zero fitted parameters. No general SST conclusion can be inferred from rejecting H1.

2 Comparator and mechanical readout

The Newtonian benchmark is

$$F_N = \frac{G m_s m_t}{r^2}, \qquad \left|\frac{dF_N}{dr}\right| = \frac{2G m_s m_t}{r^3}, \qquad \Delta F_N \simeq \left|\frac{dF_N}{dr}\right| \Delta r. \tag{1}$$

Use $G = 6.67430 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}$ as the *benchmark input only*. The published force values have a Newton-calibrated interpretation and cannot be used as an independent determination of G or $F_{\mathrm{gr}}^{\max} = c^4/(4G)$.

The reproduced static force is $4.54034e-12\,\mathrm{N}$; the force gradient is $2.162067e-10\,\mathrm{N/m}$ and the first-order modulation is $3.026893e-16\,\mathrm{N}$. The reported rounded target is $3.02 \times 10^{-16}\,\mathrm{N}$. Apparatus response, spatial geometry and detector noise must be modelled independently before inference from raw traces.

3 Euler pressure source and dimensional audit

An exact divergence-free *initial velocity field* (not a claimed stationary Euler solution) is

$$\mathbf{v}(\mathbf{x}) = \Omega \exp\!\left(-\frac{x^2+y^2+z^2}{2a^2}\right)(-y, x, 0), \quad \Omega = \frac{e^{1/2}|\mathbf{v}_{\circlearrowleft}|}{a}. \tag{2}$$

Here $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$ and $|\mathbf{v}_{\circlearrowleft}| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}}$. With $\nabla \cdot \mathbf{v} = 0$, the instantaneous Euler pressure-Poisson source is

$$\nabla^2 p = -\rho_f\, \partial_i v_j\, \partial_j v_i = 2\rho_f \Omega^2 e^{-r^2/a^2}\left(1 - \frac{x^2+y^2}{a^2}\right) \equiv S(\mathbf{x}). \tag{3}$$

The infinite-domain solution and test-volume approximation used by the code are

$$p(\mathbf{x}) = -\frac{1}{4\pi}\int_{\mathbb{R}^3} \frac{S(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}\, d^3y, \quad \mathbf{F}_p = -V_t \nabla p, \qquad V_t = \frac{4\pi a^3}{3}. \tag{4}$$

This V_t is a deliberately fixed test-volume proxy; it is *not* independently identified with the physical 100-g test mass. $[S] = \mathrm{Pa\,m^{-2}}$ and $[V_t \nabla p] = \mathrm{N}$.

The Gaussian field has zero integrated monopole $\int S\, d^3x = 0$ on the infinite domain. Thus its pressure field has no $1/r$ scalar monopole contribution, and a generic $1/r^2$ force is not expected at large separations. Both axial and equatorial orientations are tested to avoid orientation cherry-picking.

4 Explicit (candidate) mass closure

The proposed *restricted* energy-mass identification is

$$m_E = \frac{1}{2c^2}\rho_f \int_{\mathbb{R}^3} |\mathbf{v}|^2 \, d^3x = \frac{\rho_f \Omega^2 \pi^{3/2} a^5}{2c^2}. \tag{5}$$

No source mass is used to calibrate Ω: it is fully set by the canon's characteristic speed and the fixed a. The predicted equivalent mass is $1.90424e-18\,\mathrm{kg}$ and $m_E/m_s = 1.586867e-15$ (experimental source: $1.2 \times 10^{-3}\,\mathrm{kg}$). Disagreement refutes *this* postulated closure, not every possible vortex mass theory.

5 Numerics, precision and gate outcomes

The instantaneous pressure gradient is evaluated by finite-volume midpoint integration of the free-space Green function over a cube extending to $5.5a$ in every direction. Both field orientations are computed at six separations from 0.03 to 0.08 m, with grid sizes $40^3, 56^3$ and 80^3 for the full mode. Refinement is checked independent of the physical gates. Analytic $\nabla \cdot \mathbf{v} = 0$ and units are verified. A separate strict native C++ FP64 implementation checks numerical parity when an actual compiler is available; an unavailable compiler is reported as UNRESOLVED, never as a pass.

Gate	Status	Primary metric
S0	PASS	5dd2f0d5ce2df29b
S1	PASS	0.002282591
S2	PASS	3.509553e-11
S3	FAIL	1.586867e-15
S4	FAIL	2
S5	PASS	6.709169e-12
S6	UNRESOLVED	—
S7	UNRESOLVED	—
S8	UNRESOLVED	—
S9	UNRESOLVED	—

Equatorial pressure-force at 0.042 m: 0.001485011 N; axial: -0.002970023 N. Fitted $|F_p| \propto r^{-n}$ exponents: equatorial $n = 4$, axial $n = 4$. Max relative refinement changes: $3.509553e-11$, $1.591789e-11$. A violation of the mass or slope tolerance is recorded as FAIL even when earlier independent gates failed.

6 Limits, identifiability and next experiment

1. Published rounded amplitude/force numbers are not independent raw force-series data or a measurement of G.
2. The core scale $r_c = 1.40897017 \times 10^{-15}\,\mathrm{m}$ does not follow from the chosen millimetre surrogate width a. No matched-asymptotic core-to-macro calculation is provided.
3. The instantaneous Gaussian field is not a vortex knot, pressure-support closure, or an independently proven long-lived Euler solution.
4. Obtain a calibrated force or displacement sweep at *at least four separations*, repeat with two source materials, record mechanical transfer function and covariance, and withhold confirmation data from model selection.
5. Replace the Gaussian test provider with genuine PKLSA finite-core knots in a new registered protocol, deriving momentum transfer and test-object geometry rather than assuming $F = -V_t \nabla p$.

References

[1] Y. Zhu, Q. Geng, B. Xue, Y. Sun, R. Zhang, X. Chen, X. Jiang, F. Azeem, M. Tierz and W. Wan (2026), "Stopped-light-enhanced gravitational force sensing," *Nature Nanotechnology*, https://doi.org/10.1038/s41565-026-02298-8.

[2] G. K. Batchelor (1967), *An Introduction to Fluid Dynamics*, Cambridge University Press, https://doi.org/10.1017/CBO9780511800955.

[3] G. W. Gibbons (2002), "The Maximum Tension Principle in General Relativity," *Foundations of Physics* **32**, 1891–1901, https://doi.org/10.1023/A:1022370717626.

[4] H. Cavendish (1798), "Experiments to Determine the Density of the Earth," *Philosophical Transactions of the Royal Society of London* **88**, 469–526, https://doi.org/10.1098/rstl.1798.0022.